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Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights

Settings

AI-Engineering-Course / week-7 / ophelia-week7-day1.md

opheliaayamga246810-arch Add high-level explanation of LoRA b789174 · now

Preview Code Blame 27 lines (19 loc) · 997 Bytes Raw

Ophelia: Week 7, Day 1 - High Level Explanation of LoRA

High Level Explanation of LoRA

Step 1: Freeze the weights - we will not optimize them Step 2: Select some layers to target, called "Target Modules" Step 3: Create new "adaptor" matrices with lower dimensions, fewer parameters Step 4: Apply these adaptors to the Target Modules to adjust them - and these get trained.

3 Essentials Hyperparameters

1. r - The rank, or how many dimensions in the low-rank matrices.

Rule of Thumb: Start with 8, then double to 16, then 32, until diminishing returns.

2. Alpha - A scaling factor that multiplies the lower rank matrices

Rule of Thumb: Twice the value of r

3. Target Modules - Which layers of the neural network are adapted.

Rule of Thumb: Start by targeting the attention heads

Size of Weights in MB

- 13,000 - Llama 3.2 3B
- 3600 - Quantized to 8 bit
- 2200 - Quantized to 4 bits
- 8 + 3 - QLoRA with r = 32
    - The LoRA weighs in 32 bit